DIPLOMA PROGRAMME PRACTICE · HL · SET A

Chemistry / Paper 1B

Question booklet · 35 marks · 120 minutes across 1A and 1B

Answer every part. Show working, units and suitable precision. Use a calculator and a clean Chemistry data booklet. Use separate answer paper for additional working and diagrams.

QUESTION 1 [12]

Competing kinetic models

The table gives constructed initial rates for A+B→products. A calibrated optical probe tracks a product whose absorbance is linear in concentration. Temperature is constant within each run. Rate units are mol dm^{-3} s^{-1}; concentrations are mol dm^{-3}.

Run	T / K	[A]	[B]	Initial rate
1	300	0.100	0.100	0.000500
2	300	0.200	0.100	0.00200
3	300	0.200	0.200	0.00400
4	315	0.100	0.100	0.00100

(a) Determine both partial orders and the rate expression at 300 K. **[3]**

(b) Calculate k at 300 K, give its units, and predict the initial rate at [A]=0.150 and [B]=0.0800 at 300 K. **[3]**

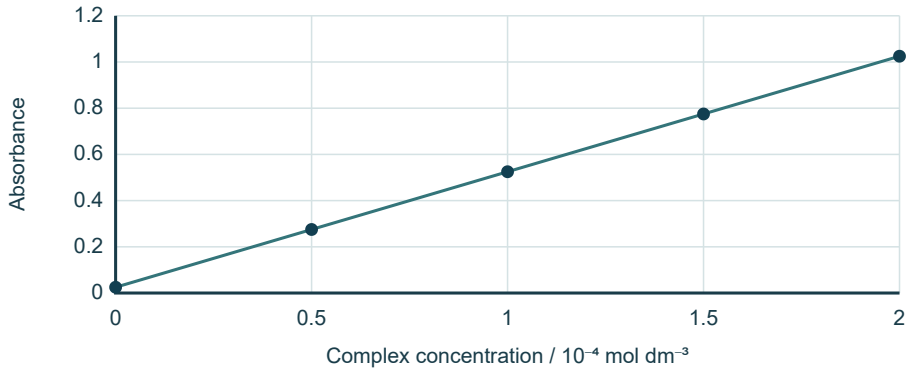
(c) Use runs 1 and 4 to estimate Ea. Use R=8.31 J K^{-1} mol^{-1} and assume the rate law and Arrhenius prefactor apply over this range. **[3]**

(d) Assess the claim that the rate law proves a single elementary three-particle collision. Suggest how to reduce bias when extracting initial rates. **[3]**

QUESTION 2 [12]

Equilibrium by absorbance

$Fe^{3+} + SCN^- \rightleftharpoons FeSCN^{2+}$. A calibration for $FeSCN^{2+}$ has intercept 0.025 and slope 5000 $dm^3\ mol^{-1}$. At the chosen wavelength other species give negligible absorbance after background correction. Initial concentrations in the final mixture are 1.000×10^{-3} mol dm^{-3} Fe^{3+} and 8.00×10^{-4} mol dm^{-3} SCN^-. Initial complex concentration is zero. The equilibrium absorbance is 0.925.



Absorbance against Complex concentration / 10^{-4} mol dm^{-3}. Data points are shown by filled circles.

(a) Use the calibration to determine the equilibrium complex concentration. Explain why the intercept must be used. **[3]**

(b) Calculate both remaining reactant concentrations and Kc using equilibrium concentrations in mol dm^{-3}. **[3]**

(c) Water doubles the volume at the same temperature. Compare the immediate Qc with Kc and predict the subsequent change in complex amount. **[3]**

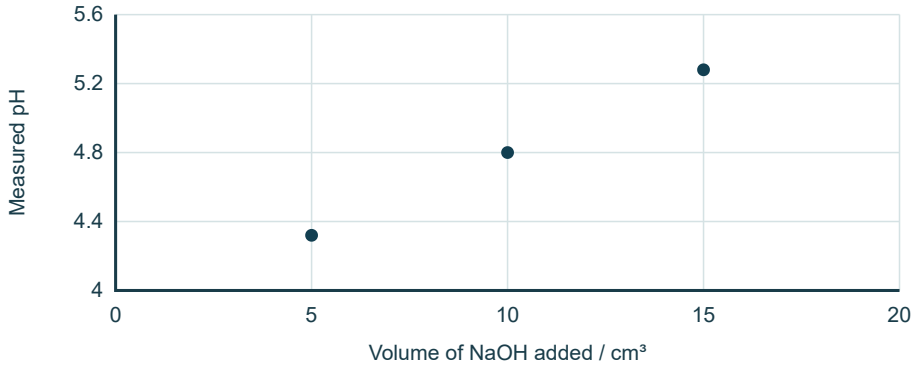
(d) Explain why a lower absorbance immediately after dilution does not, by itself, prove a shift. Design one comparison that isolates the shift. **[3]**

QUESTION 3 [11]

Locating a weak-acid endpoint

A 25.00 cm³ aliquot of HA is titrated with 0.1000 mol dm⁻³ NaOH at 298 K. Equivalence volume is 20.00 cm³, the pH there is 8.7, and the steep region spans pH 7.6–10.0. Assume volumes are additive and neglect activity effects.

NaOH added / cm³	5.00	10.00	15.00
Measured pH	4.32	4.80	5.28



Measured pH against Volume of NaOH added / cm³. Data points are shown by filled circles.

(a) Find the original analytical concentration of HA and derive pKa from the half-equivalence reading. **[3]**

(b) Using the conjugate pair ratio, predict the pH at 15.00 cm³ and compare with the reading. **[3]**

(c) Indicator M changes at pH 3.1–4.4 and N at 8.2–10.0. Choose an indicator and explain the basic equivalence pH. **[2]**

(d) At 10.00 cm³ the buffer is diluted twofold with water. Predict both its approximate pH and its buffer capacity per unit volume. State a limit to the pH prediction. **[3]**